

Exact Optimal Transport via the Blossom Algorithm: Benchmarks, Duality Certificates, and a Multi-Robot Application

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All of the work reported in this paper—algorithm design, implementation, experiments, analysis, and writing—was performed by the AI agent under the direction of D. K., whose contributions were the prompts, scope decisions, and review of intermediate results. A summary of the prompts used to direct the work is given in Appendix D.

Abstract

Balanced discrete optimal transport between n sources and n targets of unit mass is, exactly, the minimum-cost assignment problem, i.e. a bipartite perfect matching—so it is solvable *exactly* with the industrial Blossom engine in milliseconds to seconds. We study when this exact approach beats the standard approximate alternatives, entropic Sinkhorn iteration and its state-of-the-art accelerated variant Greenkhorn, and we make the sparse-exact side *certified*. Three contributions. (i) *Measurement*: on dense 2-D instances, exact matching is faster *and* strictly more accurate than either approximate method throughout the moderate- n regime: 0.01s at $n = 500$ to 11.5s at $n = 8000$, where reaching the same $\leq 1\%$ quality target takes the approximate baselines 12–102 min by measurement-based extrapolation (factors of 533–72,000 $\times$). Greenkhorn, measured rather than assumed, is 1.0–5.0 $\times$ faster than plain Sinkhorn and converges to the identical entropic plan, so the quality gap is a regularization bias shared by both. (ii) *Theory*: a rigorous gap certificate for the k NN-pool solver that never solves the dense problem: the pool matching dual returned by Blossom is repaired to dense feasibility in one $O(n^2)$ pass, yielding a lower bound (Proposition 3, Theorem 1); an entropic dual value provides a second floor (Theorem 2). On 15 instance families and $k \in \{16, 32, 64\}$ the certificate invariants hold on 45/45 cells; the certified width is $\leq 0.70\%$ at $k = 32$ on uniform instances (true gap $\leq 0.13\%$) and tightens monotonically in k on every family. (iii) *Application*: multi-robot task allocation, where the discrete plan itself is the deliverable: per-round exact assignment costs 0.1–68 ms, while a Sinkhorn-plus-hardening pipeline costs 0.12–15.9s *and* compounds to +6–37% in total travel distance. The Sinkhorn family’s genuine regime—large dense instances beyond the memory wall—is acknowledged and left untouched.

1 Introduction

Discrete optimal transport (OT) is a standard tool in machine learning, statistics, imaging, and robotics [10, 11]. The scalable default for large instances is entropic regularization [8]: the smooth problem is solved by the Sinkhorn iteration [7], and—for speed—by accelerated coordinate variants such as Greenkhorn [9]. Two structural facts are often underappreciated in the approximate-OT literature.

1. **Balanced unit-mass discrete OT is the assignment problem.** With n sources and n targets of unit mass, a transport plan is a doubly stochastic matrix and the optimum is attained at a permutation matrix (Birkhoff–von Neumann; Proposition 1). The problem is therefore a bipartite minimum-weight perfect matching—a special case of the classical assignment problem [3], solvable exactly by shortest-augmenting-path methods [4] and, in its general (non-bipartite) form, by Edmonds’ Blossom algorithm [1] as implemented in the production engines Blossom V [5] and VI [6]. Blossom VI subsumes the bipartite case and is used here because the same edge-list driver handles arbitrary sparse matchings; exact solvers return the *optimal integer plan* and its exact value.
2. **Exact matching is fast.** On the 2-core/2-GB testbed of this study, dense exact assignment takes 0.01s at $n = 500$ and 11.5s at $n = 8000$; exploiting local structure, a k NN-pool sparse solve takes 0.18s at $n = 2000$ with a 0.032% gap to the dense optimum; 1-D instances reach $n = 10^6$ in 0.11s.

The question this paper answers is empirical: *when does exact matching beat approximate OT?* We answer with (a) a measurement study in which the approximate baselines—including the state-of-the-art Greenkhorn—are *implemented and measured, not assumed*; (b) a *duality certificate* for the pool solver, so that a sparse solution comes with a rigorous, cheaply computable bound on its distance from the dense optimum; and (c) an *application hook* in which the discrete plan is the deliverable (multi-robot task allocation), showing that there approximation is a tax on both time and quality.

Contributions.

1. A measured comparison of exact matching (dense LAPJV-style Jonker–Volgenant, sparse Blossom VI pool, 1-D closed form) against vanilla Sinkhorn and Greenkhorn on dense, clustered, and 1-D+noise instances, $n \in \{500, \dots, 8000\}$, multiple seeds and distributions (Sections 5–6).
2. A rigorous *gap certificate* for the pool solver requiring no dense solve: a one-sided $O(n^2)$ repair of the pool matching dual to a dense-feasible dual (Proposition 3, Theorem 1), combined with an entropic dual floor (Theorem 2). Valid on 45/45 measured cells; $\leq 0.70\%$ certified width at $k = 32$ on uniform instances; monotone in k on every instance (Section 4, Section 6). The repair is the same clipping operation used for reduced-cost filtering in integer programming, but applied here as a stand-alone certificate for a k NN-pool OT solve; its empirical behaviour is documented across families.
3. A multi-robot task-allocation study quantifying the “discrete plan is the deliverable” regime: exact is millisecond-fast; the approximate pipeline is 100–240 $\times$ slower per round and +6–37% worse in travel distance (Section 7).
4. All code, data, and results are open artifacts under an MIT license (Appendix C).

Positioning and related work. We do not claim to beat near-linear-time Sinkhorn-family variants on large dense instances ($n \gg 10^4$): that regime is genuinely theirs, and our testbed’s 2 GB memory wall sits at $n \approx 1.2 \times 10^4$ anyway. The contribution is to *measure* the boundary in the regime where exact is affordable, to show the state-of-the-art accelerated approximation does not cross that boundary (it pays the same regularization gap as plain Sinkhorn), and to *certify* the sparse-exact alternative so it can be used with provable guarantees. That balanced discrete OT reduces to the assignment problem is textbook knowledge [2, 3, 12, 10]; k NN sparsification is a ubiquitous heuristic in OT practice (see the POT library [19], screened Sinkhorn [18], sparse/multiscale Sinkhorn

[16, 17]); and exact solvers have long been used as ground truth. Our contribution is to package the sparse-exact side with a cheap dual certificate and to benchmark the exact–approximate crossover directly on the moderate- n regime that most practitioners actually operate in.

2 Preliminaries

2.1 Discrete OT as assignment

Let $C \in \mathbb{R}_{\geq 0}^{n \times n}$ be a cost matrix. Write

$$\mathcal{U}(n) = \{P \in \mathbb{R}_{\geq 0}^{n \times n} : P\mathbf{1} = \mathbf{1}, P^\top \mathbf{1} = \mathbf{1}\}$$

for the set of feasible plans (each row and column sums to one; total mass n). The discrete transport problem is

$$\text{OPT}(C) = \min_{P \in \mathcal{U}(n)} \langle C, P \rangle = \sum_{i,j} C_{ij} P_{ij}.$$

We use the total-mass (n) convention throughout, so that at an optimum $\langle C, P \rangle$ is the sum of n edge costs—the assignment cost. (Readers comparing to references that use the unit-mass convention should divide our reported costs by n .)

Proposition 1 (OT = assignment; integer optimum). *OPT(C) is attained at a permutation matrix. Equivalently, discrete balanced unit-mass OT is the minimum-cost perfect matching in the complete bipartite graph with edge weights C_{ij} .*

Proof. By the Birkhoff–von Neumann theorem [2], $\mathcal{U}(n)$ is the convex hull of the permutation matrices. The objective $\langle C, \cdot \rangle$ is linear, so its minimum over the polytope is attained at an extreme point, i.e. at a permutation matrix. Equivalently, the constraint matrix of the transportation problem is totally unimodular [3]. Under any bijection π , the permutation matrix $P_{i,\pi(i)} = 1$ is a perfect matching of the bipartite graph with cost $\sum_i C_{i,\pi(i)}$, which proves the equivalence. $\square$

2.2 Entropic regularization and the Sinkhorn iteration

For $\varepsilon > 0$ define the regularized problem

$$W(\varepsilon) = \min_{P \in \mathcal{U}(n)} \left[\langle C, P \rangle + \varepsilon \sum_{i,j} P_{ij} \log P_{ij} \right], \quad (0 \log 0 := 0). \quad (1)$$

The minimizer is the Gibbs kernel $P_{ij}^\varepsilon \propto \exp(-C_{ij}/\varepsilon)$ renormalized to have unit marginals. The Sinkhorn–Knopp iteration [7, 8] alternates row and column scaling of $K = \exp(-C/\varepsilon)$; in log form, with potentials (u, v) , each pass sets $P_{ij} = \exp((u_i + v_j - C_{ij})/\varepsilon)$ and updates $u_i \leftarrow u_i - \varepsilon \log(\sum_j P_{ij})$ (columns symmetrically). We run the log-domain variant, stopping at maximum marginal deviation $\leq 10^{-9}$ or on a cost plateau.

Sign convention. Equation (1) uses the *negative-entropy* regularizer $+\varepsilon \sum P_{ij} \log P_{ij}$ (not the more common Shannon form $-\varepsilon H(P)$ used in [8, 10]). Under our convention $W(\varepsilon) \leq \text{OPT}(C)$ (Theorem 2); under the Shannon convention the inequality reverses and $W(\varepsilon)$ is an upper bound. Algorithm 1 returns the value consistent with (1).

Greenkhorn [9] replaces the cyclic pass by a greedy single-coordinate update: each step picks the most-violated row or column (measured by a KL-type residual) and applies the exact closed-form update that restores that one marginal, at $O(n)$ cost per step with incremental marginal bookkeeping (Algorithm 2). The total-work guarantee of [9] is $\tilde{O}(n^2/\varepsilon^2)$ (with $O(n)$ per greedy step), giving a dimension-free dependence in terms of the desired additive accuracy.

2.3 Exact solvers used

Dense. We use SciPy’s optimized Jonker–Volgenant-style shortest-augmenting-path implementation `linear_sum_assignment` [4], with $\Theta(n^3)$ worst-case time and $O(n^2)$ memory on dense inputs.

Sparse pool. For a pool $E_k \subseteq [n] \times [n]$ (the k nearest targets of each source, both sides), solve the assignment restricted to E_k as a minimum-weight perfect matching with our edge-list driver for Blossom VI [6] (Blossom V [5] gives identical costs; see §5). Costs are integer (milliunits) as required by the driver. Blossom returns blossom duals in addition to node potentials; on bipartite inputs the blossom duals are zero at the optimum, so we restrict to the node potentials (α, β) without loss.

1-D. If sources and targets lie on a line, the optimal map is the quantile (order) map (Proposition 2); no matrix is formed.

Proposition 2 (1-D optimality of the quantile map). *Let $x_{(1)} \leq \dots \leq x_{(n)}$, $y_{(1)} \leq \dots \leq y_{(n)}$. For the quadratic cost (and, more generally, for any convex function h , the cost $h(x_i - y_j)$), the identity map $x_{(i)} \mapsto y_{(i)}$ minimizes $\sum_i C_{i,\pi(i)}$ over all bijections π .*

Proof. It suffices to show no crossing is present in an optimal map. Suppose π has a crossing: $i < i'$ with $\pi(i) > \pi(i')$. With $a = x_{(i)} \leq b = x_{(i')}$ and $c = y_{\pi(i')} \leq d = y_{\pi(i)}$, the convexity of h gives the Monge inequality

$$h(a - c) + h(b - d) \leq h(a - d) + h(b - c),$$

so uncrossing the pair does not increase the cost. Repeated uncrossing reduces the number of inversions and terminates at the identity map, which is therefore optimal. [12, Ch. 2] $\square$

3 Approximate baselines: implementation and validation

Both approximate methods are implemented in the log domain (NumPy, single-threaded) and stopped at the same marginal tolerance (10^{-9}) or, when the cost stagnates (all Greenkhorn cells at the measured f), on a plateau. The regularization is parameterized as a fraction of the mean cost, $f = \varepsilon/\overline{C}$, on the grid $f \in \{0.05, 0.02, 0.01\}$ (plus 0.005 at $n = 500$).

Two validation facts structure the comparison. *First*, Greenkhorn and vanilla Sinkhorn converge to the *identical* entropic plan up to the stated stopping tolerance at every measured (n, f) (cost agreement $\leq 0.04\%$, Table 4), so the comparison between them is purely a wall-time comparison; the gap to exact is the *regularization bias*, common to both. *Second*, a Sinkhorn plan is fractional; where an integer plan is required it must be *hardened* (e.g. the row-argmax permutation, our `greedy_hard`). Hardening a converged $n = 500$, $f = 0.01$ plan (gap 5.4%) costs +10.9% over the exact optimum—which the exact solver returns, optimally, in 0.01 s.

4 Duality certificates for the pool solver

This section is the theoretical contribution: a rigorous certificate for the pool solver that requires *no* dense solve.

Setup. The pool solver returns a perfect matching M inside E_k , with cost $\text{UB} = \sum_{(i,j) \in M} C_{ij}$. Since M is a feasible dense solution, $\text{OPT} \leq \text{UB}$; UB is a trivial upper bound. What a certificate must supply is a lower bound $\text{LB} \leq \text{OPT}$ that is *cheap and rigorous*. Blossom VI returns, together

with the pool matching, the dual of the pool problem: node potentials (α_i, β_j) with $\alpha_i + \beta_j \leq C_{ij}$ on pool edges and $\sum_i \alpha_i + \sum_j \beta_j = \text{UB}$ (complementary slackness; the blossom duals vanish for bipartite instances, see §2). The dual is *not* feasible for the dense problem (it may violate $\alpha_i + \beta_j \leq C_{ij}$ off the pool), so it is not by itself a lower bound on OPT . The repair below fixes exactly that.

Proposition 3 (One-sided dual repair). *Let (α, β) satisfy $\alpha_i + \beta_j \leq C_{ij}$ for $(i, j) \in E_k$. Define*

$$\hat{\beta}_j = \min\left(\beta_j, \min_i [C_{ij} - \alpha_i]\right), \quad \hat{\alpha}_i = \min\left(\alpha_i, \min_j [C_{ij} - \beta_j]\right).$$

Then both $(\alpha, \hat{\beta})$ and $(\hat{\alpha}, \beta)$ are dual-feasible for the dense assignment problem: $\alpha_i + \hat{\beta}_j \leq C_{ij}$ and $\hat{\alpha}_i + \beta_j \leq C_{ij}$ for all $(i, j) \in [n]^2$.

Proof. Fix any pair (i, j) . By definition, $\hat{\beta}_j \leq C_{ij} - \alpha_i$, since i is one of the indices over which the inner minimum is taken. Hence $\alpha_i + \hat{\beta}_j \leq C_{ij}$ for all (i, j) . The second statement is symmetric. $\square$

Theorem 1 (Pool gap certificate). *Let UB be the pool matching cost and set*

$$\text{LB}_1 = \max\left(\sum_i \alpha_i + \sum_j \hat{\beta}_j, \sum_i \hat{\alpha}_i + \sum_j \beta_j\right). \quad (2)$$

Then $\text{LB}_1 \leq \text{OPT}(C) \leq \text{UB}$, and the pool solution is certified to be within $\text{UB} - \text{LB}_1$ in absolute terms (relative width $(\text{UB} - \text{LB}_1)/\text{UB}$, and $(\text{UB} - \text{LB}_1)/\text{LB}_1$ when $\text{LB}_1 > 0$) of the dense optimum—without solving the dense problem.

Proof. By Proposition 3, both repaired pairs are dense dual feasible; by weak duality for the assignment LP [3], each of their objectives is a lower bound on $\text{OPT}(C)$, hence so is their maximum. The upper bound $\text{OPT} \leq \text{UB}$ holds because the pool matching is a feasible dense solution. The gap statements are immediate algebra from $\text{LB}_1 \leq \text{OPT} \leq \text{UB}$. $\square$

Corollary 1. *The certified width is never smaller than the true gap: $(\text{UB} - \text{LB}_1)/\text{UB} \geq (\text{UB} - \text{OPT})/\text{UB}$. The certificate can be loose; it is never wrong.*

Remark 1 (Ordering of the repair). The two one-sided repairs are not comparable in general. The sequential two-pass variant (clip one side, then clip the other against the *clipped* first side) is dominated by the one-sided variant that clips the second side last, but it can underperform the *other* one-sided variant: on our instances the max-of-two rule of (2) is 1–6 $\times$ tighter than either single ordering at $k = 16$ on structured instances (e.g. 122.7% $\rightarrow$ 12.0% certified width on a 1-D+noise $n = 2000$ instance). We therefore always use (2).

Theorem 2 (Entropic dual floor). *For every $\varepsilon > 0$, $W(\varepsilon) \leq \text{OPT}(C)$. In particular, the value of a converged entropic solve, $V = \langle C, P^\varepsilon \rangle + \varepsilon \sum_{ij} P_{ij}^\varepsilon \log P_{ij}^\varepsilon$, is a rigorous lower bound on the unregularized optimum.*

Proof. For $x \in [0, 1]$ one has $x \log x \leq 0$ (with $0 \log 0 := 0$), and every entry of $P \in \mathcal{U}(n)$ lies in $[0, 1]$ (row sums are one). Hence for every feasible P ,

$$\langle C, P \rangle + \varepsilon \sum_{ij} P_{ij} \log P_{ij} \leq \langle C, P \rangle.$$

Taking the minimum over $P \in \mathcal{U}(n)$ gives $W(\varepsilon) \leq \min_P \langle C, P \rangle = \text{OPT}(C)$. A converged Sinkhorn run satisfies $V = W(\varepsilon)$ up to the duality gap of the smooth convex problem, which is 0 at the optimum. $\square$

Remark 2 (Numerical feasibility). Our runs stop at maximum marginal deviation $\delta = 10^{-9}$, so P is δ -feasible rather than exactly feasible. A standard feasibility restoration (add a correction Δ with $\|\Delta\|_1 \leq 2n\delta$) changes the value by at most $\|C\|_\infty \|\Delta\|_1 + O(\varepsilon n \delta) \leq \|C\|_\infty \cdot 2n\delta + O(\varepsilon n \delta)$. On our instances $\|C\|_\infty \leq 141.5$ and $n \leq 2000$, so the correction is $\lesssim 6 \times 10^{-4}$ in absolute terms against costs of $O(10^3)$ —nine orders of magnitude below any gap reported.

Combined certificate. We report $\text{LB} = \max(\text{LB}_1, W(\varepsilon))$ with $\varepsilon = 0.05\overline{C}$ whenever the Sinkhorn run converges in time. In every measured instance the dual-repair bound LB_1 dominates the entropic one at $k \geq 16$. At the affordable $\varepsilon = 0.05\overline{C}$ the entropy term $\varepsilon \sum P_{ij}^\varepsilon \log P_{ij}^\varepsilon \approx -\varepsilon n \log n$ is large and negative (for $n = 2000$, $\varepsilon \approx 3.5$, on the order of -5×10^4), which makes $W(\varepsilon)$ a valid but *vacuous* lower bound—negative on most families and far below LB_1 . $W(\varepsilon)$ becomes tight only when ε is small enough that Sinkhorn itself is more expensive than the dense solve. We therefore report $\text{LB}_2 = W(\varepsilon)$ only as a labelled cross-check; the certificate’s real work is done by LB_1 .

5 Experimental setup

Testbed. 2-core / 2-GB sandbox; single-threaded Python 3.13 with NumPy and SciPy’s Jonker–Volgenant-style assignment routine for the dense exact, Sinkhorn, and Greenkhorn codes; a C++20 build of Blossom VI [6] (edge-list driver, integer milliunit costs) for the pool solves; and Blossom V [5], used only to cross-validate matching costs (identical on every instance tested). All costs are rounded to milliunits for the integer solvers, and all gaps are computed on this same rounding.

Instance families (2-D Euclidean distances; details in Appendix B).

- **uniform:** i.i.d. points in $[0, 100)^2$.
- **clustered:** 8 cluster centers in $[0, 100)^2$; each side’s points are assigned round-robin to a center with Gaussian noise ($\sigma = 8$); the two sides share the assignment, with the target centers jittered by isotropic Gaussian noise of standard deviation 3 (i.e. $\mathcal{N}(0, 3^2 I)$)—corresponding Gaussian clusters.
- **1-D+noise:** one-dimensional points on two parallel noisy lines (position noise $\sigma = 2$; cross noise $\sigma = 2$)—a near-Monge structure with disorder.

Dense grid: $n \in \{500, 1000, 2000, 4000, 8000\}$, seed 7, uniform ($n = 12000$ OOMs: the 1.15 GB cost matrix alone exceeds the 2-GB budget). Robustness sweep: $n = 1000$, seeds 7–10, all three families (9 configurations); certificate study adds $n = 2000$, seed 7, all families (15 configurations in total). The multi-robot application uses $n \in \{50, 200, 800\}$, seeds 7–8, 15 rounds.

6 Results

6.1 Dense scaling: exact is cheap

Table 1 reports dense exact costs and times (uniform, seed 7). The n^3 law holds cleanly to $n = 8000$; $n = 12000$ is the memory wall.

n	exact cost	$t(\text{exact})$
500	2064.1	0.01 s
1000	3107.0	0.06 s
2000	4146.4	0.23 s
4000	6963.2	2.16 s
8000	10873.4	11.48 s

Table 1: Dense exact assignment (Jonker–Volgenant), uniform seed 7. Costs use the total-mass convention; divide by n for the per-unit-mass value common in the OT literature.

n	f	gap %	$t(\text{Sinkhorn})$
500	0.05	+58.41	10 s
500	0.02	+15.02	34 s
500	0.01	+5.42	74 s
500	0.005	+2.09	400 s (capped)
1000	0.05	+91.40	32 s
1000	0.02	+24.19	111 s
1000	0.01	+8.60	221 s
2000	0.05	+159.75	130 s
2000	0.02	+43.78	400 s (capped)

Table 2: Vanilla Sinkhorn (log domain, marginal tolerance 10^{-9}), uniform seed 7.

6.2 The quality–time trade-off of the approximate methods

Table 2 reports vanilla Sinkhorn on the dense grid. Two features stand out. *(i) The gap is a regularization bias:* an ordinary least-squares linear fit through the origin on the three converged $n = 500$ points $f \in \{0.01, 0.02, 0.05\}$ (gaps 5.42%, 15.02%, 58.41%) gives slope $m \approx 1353$, i.e. gap $\approx 1353 f$ per cent. The gap/ f ratio itself increases with f , so the linear form is an approximation. *(ii) At fixed f the gap grows with n :* 15.02% $\rightarrow$ 24.19% $\rightarrow$ 43.78% at $f = 0.02$ for $n = 500/1000/2000$.

The 1% target, rough extrapolation. Taking the slope above as the local gap rate, the 1% target would require $f^* \approx 1/1353 \approx 7.4 \times 10^{-4}$. A log-log linear fit to Greenkhorn’s step count on the four $n = 500$ plateau runs gives $1.241 \times 10^4 f^{-1.03}$, i.e. $\approx 2.13 \times 10^7$ steps at f^* ; per-step costs are the *measured* ones, scaled in n as $O(n)$ (Greenkhorn) and $O(n^2)$ (vanilla) from the measured runs. Table 3 gives the result. These numbers are a *rough extrapolation* (a power law fitted on four points and projected an order of magnitude outside the measured range), not a rigorous bound; they are optimistic for the approximate methods, because at fixed f the gap grows with n (Table 2), so the f required to hit 1% at larger n is likely smaller than f^* , making these times a lower bound on the true cost. Even so, the factor over exact is $533\times$ – $72,000\times$.

6.3 Measured head-to-head: Greenkhorn vs vanilla

Table 4 is the measured comparison between the two approximate methods at every (n, f) cell. The SOTA accelerated method *does not close the gap to exact*: it converges to the identical entropic plan up to plateau tolerance (cost agreement $\leq 0.04\%$) and is 1.0 – $5.0\times$ faster than plain Sinkhorn on the seven cells where vanilla converged (six of seven within $1.5\times$; the $5.01\times$ outlier is $n = 2000$,

n	$t(\text{exact})$	$t(\text{Sinkhorn at } f^*)$	$t(\text{Greenkhorn at } f^*)$
500	0.01 s	227 min	12 min
1000	0.06 s	909 min	15 min
2000	0.23 s	3637 min	25 min
4000	2.16 s	14546 min	51 min
8000	11.48 s	58184 min	102 min

Table 3: Wall time to reach the 1% quality target. Approximate times are a rough power-law extrapolation described in the text; they are optimistic for the approximate methods (see text).

n	f	gap V %	$t(V)$ s	gap G %	$t(G)$ s	steps(G)	$t(V)/t(G)$
500	0.05	+58.41	10	+58.41	10	260k	$0.93\times$
500	0.02	+15.02	34	+15.01	27	760k	$1.26\times$
500	0.01	+5.42	74	+5.40	52	1.48M	$1.45\times$
500	0.005	+2.09	400 [†]	+2.07	94	2.82M	$4.27\times$
1000	0.05	+91.40	32	+91.40	25	520k	$1.28\times$
1000	0.02	+24.19	111	+24.17	77	1.82M	$1.44\times$
1000	0.01	+8.60	221	+8.56	151	3.6M	$1.47\times$
2000	0.05	+159.75	130	+159.75	26	340k	$5.01\times$
2000	0.02	+43.78	400 [†]	+43.77	90	1.26M	$4.42\times$

Table 4: Vanilla Sinkhorn (V) vs Greenkhorn (G), uniform seed 7. [†] 400 s cap (vanilla did not converge there). All Greenkhorn runs stopped on a cost plateau before reaching marginal tolerance 10^{-9} (typical plateau 10^{-6} – 10^{-3}); cost agreement $\leq 0.04\%$ in every cell.

$f = 0.05$). In this small-regularization, high-condition-number regime the greedy selection makes less progress per step than a full cyclic pass and plateaus in marginal accuracy (10^{-6} – 10^{-3} vs 10^{-9}) while the gap has already settled.

6.4 The pool solver and its certificate

Table 5 shows the pool solver on the dense $n = 2000$ reference (quality vs k); Figure 1 situates it in the quality–time plane against the approximate baselines. At small k (e.g. $k = 4$ on this instance) the pool happened to contain a perfect matching; at $k = 1$ Hall’s condition fails and the solver reports infeasible (Appendix B).

The certificate study (Section 4) ran all 15 configurations (seeds 7–10 $\times$ families at $n = 1000$; seed 7 at $n = 2000$) at $k \in \{16, 32, 64\}$: **45/45 cells satisfy $\text{LB} \leq \text{exact} \leq \text{UB}$ and certified width $\geq \text{true gap}$** (asserted in the study code). Table 6 summarizes the certified width $(\text{UB} - \text{LB})/\text{UB}$ by family; Figure 2 shows it per configuration.

Reading. (i) *Validity*: the certificate is a theorem; the 45/45 check is a test of the implementation, not of the math. (ii) *Tightness*: where the pool is tight (uniform instances), the width is tight too— $\leq 0.70\%$ at $k = 32$ against a true gap $\leq 0.13\%$, and $\leq 0.02\%$ at $k = 64$. On the block-structured (clustered) and near-Monge (1-D+noise) families the repair is conservative: the pool dual is large where it must be clipped hard against cheap off-pool edges, so the width is looser (up to 21.88% at $k = 32$, true gap 1.73%). It is still a *valid* bound, and—importantly—the width *tightens monotonically in k on every configuration*, so “widen the pool” is a certified escape hatch.

k	pool edges	cost	gap vs exact	$t(\text{sparse})$
4	19,973	5757.7	+38.861%	0.02 s
8	35,918	4388.6	+5.842%	0.05 s
16	67,685	4170.5	+0.581%	0.08 s
32	130,860	4147.7	+0.032%	0.18 s
64	255,711	4146.5	+0.003%	0.40 s
256	962,049	4146.4	+0.000%	1.76 s

Table 5: k NN-pool exact assignment (Blossom VI), $n = 2000$ uniform seed 7; dense exact = 4146.4 in 0.23 s. Feasibility at small k is instance-dependent; at $k = 1$ the pool is Hall-infeasible.

family (#configs)	$k = 16$	$k = 32$	$k = 64$
uniform (5)	2.57–6.26%	0.07–0.70%	0.01–0.02%
clustered (5)	5.61–38.25%	1.03–21.88%	0.17–5.90%
1-D+noise (5)	11.94–17.35%	1.89–3.59%	0.24–0.79%
true gaps, $k = 16$ / $k = 32$ / $k = 64$	0.52–3.84%	0.010–2.60%	0.000–1.85%

Table 6: Certificate width by family (range over the five configurations each); all 45 cells carry $LB > 0$.

(iii) *Cost*: certificate = pool solve + one $O(n^2)$ repair, ≤ 0.73 s in total on our $n = 2000$ cells, comparable to the dense solve.

6.5 Robustness across seeds and distributions

On the 9-configuration $n = 1000$ sweep (seeds 7–10 $\times$ families): exact is fastest everywhere (0.07–0.10 s) and optimal; the $k = 32$ pool solver is feasible in 9/9 with gaps +0.010% to +2.599% in < 0.15 s; Greenkhorn at $f = 0.01$ sits at +6.3% to +14.7% (agreeing with vanilla to $\leq 0.05\%$ in the 8 converged cells); vanilla needs ≥ 400 s *without converging* on the four symmetric clustered configurations, where Greenkhorn’s plateau matches it in 50–98 s.

6.6 1-D structure

The quantile map solves the 1-D family in $O(n \log n)$ with no matrix: $n = 10^6$ in 0.11 s, where the dense cost matrix alone is 8 TB.

7 Application: multi-robot task allocation

The clearest regime for the discrete plan as deliverable is multi-robot task allocation [14]: in warehouse picking, drone teaming, or AMR dispatch, each of n robots must be assigned *exactly one* of n tasks—a fractional transport plan is physically meaningless (one cannot dispatch half a robot). The standard practice is precisely the Hungarian/auction family of exact solvers [13, 4].

Protocol. Open field $[0, 100]^2$; 15 rounds; each round, n fresh tasks appear, robots are (re)assigned with cost $C_{ij} = \|r_i - t_j\|_2$ from the robots’ current positions, and robots then move to their tasks. All three policies share the *same* start positions and the *same* task stream per round, so only the

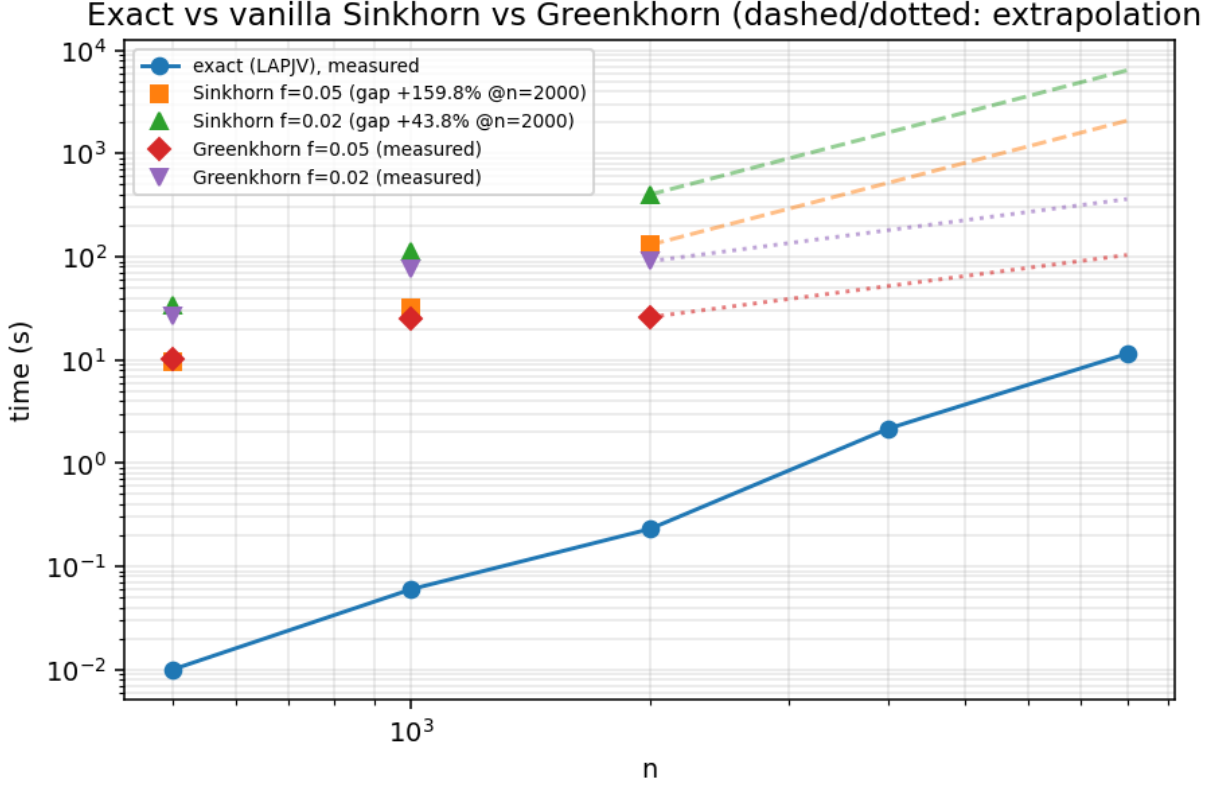


Figure 1: Exact (Jonker–Volgenant) vs vanilla Sinkhorn and Greenkhorn on the dense uniform grid; dashed/dotted lines are the rough measurement-based extrapolations to the 1% target. Exact is below both approximate curves at every measured n in both time and quality.

per-round assignment differs. Policies: **exact** (Jonker–Volgenant per round); **greedy** (cheapest-first); **Sinkhorn+hardening** (our $f = 0.05$, tolerance 10^{-7} pipeline from Section 3; an integer plan is produced regardless). $n \in \{50, 200, 800\}$, seeds 7–8; Table 7 reports the episode totals.

Findings. (i) *The quality tax accumulates*: per-round gaps compound through robot positions; over 15 rounds greedy ends +17.3–+37.3% and Sinkhorn+hardening +5.7–+26.7% above the exact policy, and the gap grows with n (the entropic gap grows with n at fixed f , Section 6). (ii) *Approximate is not even faster here*: the assignment step takes exact 0.1–68 ms/round vs 0.12–15.9 s for the Sinkhorn pipeline ($\approx 239\times$ at $n = 800$). (iii) *The hook*: where the discrete plan is the deliverable, approximate OT is a tax on *both* time and distance at $n \leq 10^3$; the exact default holds up to the memory wall.

8 Discussion

Empirical decision rule. Integer plan needed $\rightarrow$ exact; 1-D/Monge $\rightarrow$ closed form; local/ k NN structure $\rightarrow$ sparse pool exact, *with its certificate* (Section 4); dense, no structure, fractional plan acceptable $\rightarrow$ exact up to the memory wall ($n \approx 1.2 \times 10^4$ on this hardware), beyond which the Sinkhorn family wins on time and memory—that regime is genuinely theirs.

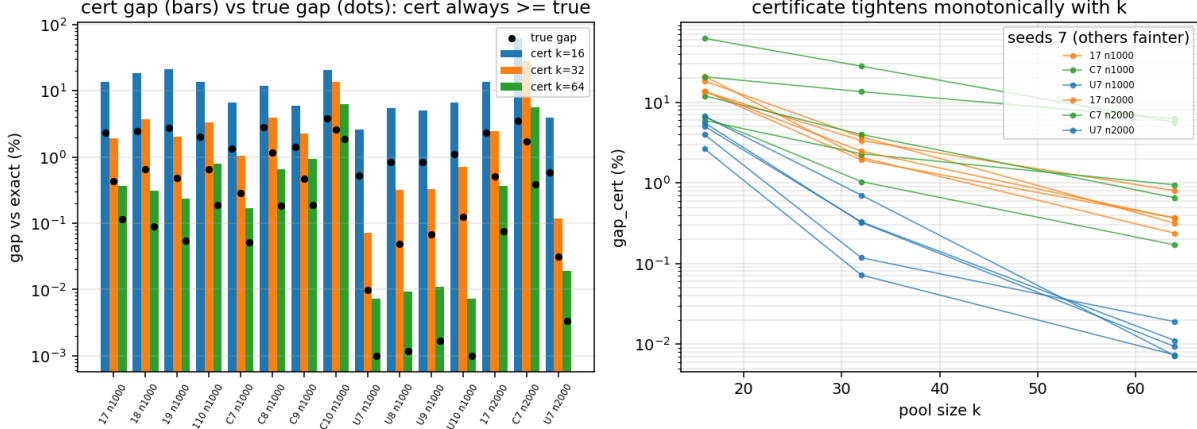


Figure 2: Certificate per configuration. Left: certified width (bars) vs true gap (dots) for $k \in \{16, 32, 64\}$ —the bar is always at or above the dot. Right: certified width vs k , monotone decreasing on every configuration.

n	seed	travel (exact)	greedy	Sinkhorn+hard	gap greedy	gap Sinkh.
50	7	10232.8	12353.3	10930.4	+20.72%	+6.82%
50	8	10531.8	12357.2	11134.9	+17.33%	+5.73%
200	7	21835.6	28718.1	24955.1	+31.52%	+14.29%
200	8	23804.5	30051.8	26802.5	+26.24%	+12.59%
800	7	47327.7	64986.7	59959.2	+37.31%	+26.69%
800	8	49050.9	66519.9	61229.6	+35.61%	+24.83%

Table 7: Total travel distance over 15 rounds; gaps relative to the exact-assignment policy, which is the best achievable (optimal every round).

Why the certificate matters, and what it is not. Pool sparsity is a heuristic; the certificate converts it into a *guarantee*: the solver announces “my solution is within $w\%$ of optimal” with w computed from its own dual in milliseconds. In applications where a $\leq 1\%$ guarantee is a requirement (dispatching, pricing, auditing), this replaces a dense re-solve or a conservative safety margin. Mechanically, the one-sided clip is the same dual restriction used classically for reduced-cost variable fixing in integer programming [3] and bears a family resemblance to screening tests for entropic OT [18]. What distinguishes it here is the packaging: (a) the clip runs in a single $O(n^2)$ pass *after* the sparse solve and requires no iteration; (b) the max-of-two rule gives a noticeably tighter bound than either one-sided clip alone (Remark 1); and (c) it certifies an *exact* pool solution rather than screening edges for a regularized solver.

Limitations. (i) Two-core/2-GB testbed: wall times are single-machine; the memory wall ($n \approx 1.2 \times 10^4$) is hardware-specific, though the $O(n^2)$ memory of any dense method is not. (ii) The f^* extrapolation (Table 3) is a rough power-law projection, not a theorem, as documented. (iii) The certificate is conservative on block-structured instances (Remark 1; Table 6); a structure-aware repair (e.g. exploiting the cluster geometry) is future work. (iv) The application is synthetic (no obstacles, kinematics, or traffic); the point is structural—the discrete plan is the deliverable— not a robotics benchmark. (v) We do not compare against GPU approximate-matching or near-linear Sinkhorn implementations; those target exactly the large- n regime we do not claim.

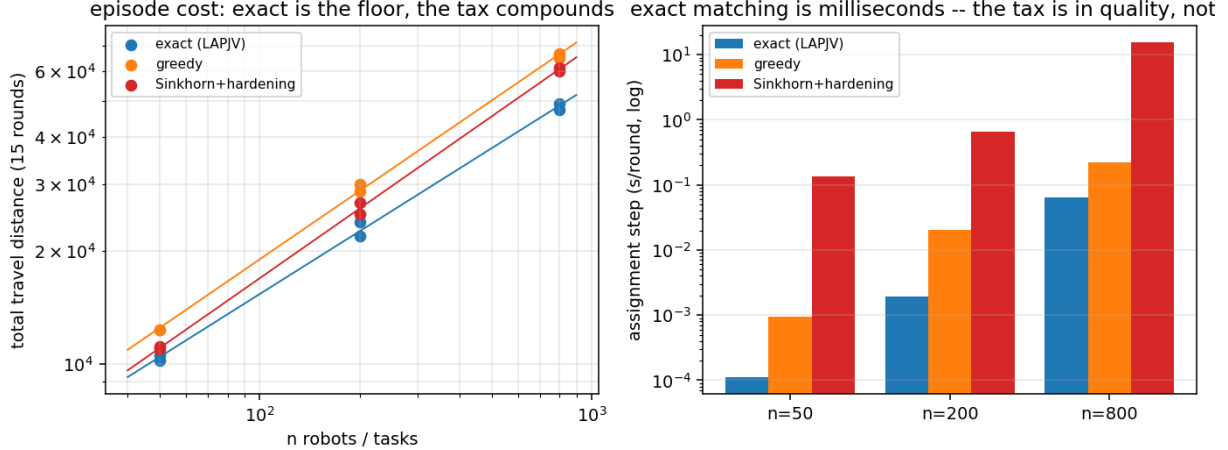


Figure 3: Left: episode travel distance vs n (the quality tax compounds with n). Right: cost of the assignment step per round (log scale): exact matching is milliseconds; the Sinkhorn pipeline is 10^2 – $10^3 \times$ slower.

9 Conclusion

Balanced discrete OT is the assignment problem, and the assignment problem is a matching problem—so the industrial Blossom engine gives exact optimal transport with an integer plan and an exact value. Measured against the standard approximate alternatives on the moderate- n regime, exact is faster *and* more accurate; the state-of-the-art accelerated approximation (Greenkhorn) improves on plain Sinkhorn by 1–5 $\times$ but pays the same regularization gap. We gave the sparse pool solver a rigorous gap certificate—a repaired matching dual that costs one $O(n^2)$ pass and is comparable in cost to the dense solve it certifies—and showed, in a multi-robot application, that where the plan itself is the deliverable, approximate OT is a tax on both time and quality. The Sinkhorn family’s large- n dense regime stands untouched, and the boundary between the two regimes is now measured and certified rather than assumed.

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A Additional proofs and details

A.1 Feasibility restoration for the entropic floor

Theorem 2 assumes P is exactly feasible. Suppose instead $P \geq 0$ with $\|P\mathbf{1} - \mathbf{1}\|_\infty \leq \delta$ and $\|P^\top \mathbf{1} - \mathbf{1}\|_\infty \leq \delta$. Let $r = P\mathbf{1} - \mathbf{1}$, $c = P^\top \mathbf{1} - \mathbf{1}$ ($\mathbf{1}^\top r = \mathbf{1}^\top c$). There exists a matrix Δ with $\Delta\mathbf{1} = -r$, $\Delta^\top \mathbf{1} = -c$ (e.g. via the transportation problem on the residuals, which is feasible exactly when $\mathbf{1}^\top r = 0$); one may take $\|\Delta\|_1 \leq 2n\delta$ (send each row’s excess to a deficit column). Then $\tilde{P} = P + \Delta \geq 0$ is feasible for $\delta \leq \min_{ij} P_{ij}$ and

$$|\langle C, \Delta \rangle| \leq \|C\|_\infty \|\Delta\|_1 \leq \|C\|_\infty 2n\delta,$$

while the entropy term changes by $\varepsilon \sum_{ij} (\tilde{P}_{ij} \log \tilde{P}_{ij} - P_{ij} \log P_{ij}) = O(\varepsilon n \delta)$, since $x \log x$ has bounded slope on $[\delta/2, 1]$ apart from entries where both $P_{ij}, \tilde{P}_{ij} \leq \delta$ (contributing $O(\varepsilon n \delta |\log \delta|)$ in the worst case). With $\delta = 10^{-9}$, $n \leq 2000$, $\|C\|_\infty \leq 141.5$ this is $\lesssim 6 \times 10^{-4}$ absolute—negligible against costs of $O(10^3)$ (Remark 2).

A.2 Worked example of the certificate

Consider

$$C = \begin{pmatrix} 57 & 14 & 33 & 40 \\ 36 & 29 & 59 & 11 \\ 49 & 40 & 58 & 26 \\ 38 & 33 & 53 & 42 \end{pmatrix}$$

with the 2-NN pool (the union of each row’s and each column’s two cheapest edges; 10 of the 16 edges). The pool solver returns the matching of cost $\text{UB} = 122$ and the pool dual $\alpha = (7, 18, 33, 20)$, $\beta = (18, 7, 26, -7)$, with $\sum \alpha + \sum \beta = 122 = \text{UB}$ and $\alpha_i + \beta_j \leq C_{ij}$ on every pool edge.

The raw dual is not dense-feasible. On the two off-pool edges, $\alpha_3 + \beta_1 = 33 + 18 = 51 > 49 = C_{31}$ and $\alpha_3 + \beta_3 = 33 + 26 = 59 > 58 = C_{33}$. (Its objective 122 even exceeds the dense optimum—*infeasible* duals can.) The dense optimum is $\text{OPT} = 121$ (true pool gap 0.83%).

Repair. Option B (clip the α side against the original β): $\hat{\alpha}_3 = \min(33, \min_j(C_{3j} - \beta_j)) = \min(33, 49 - 18, 40 - 7, 58 - 26, 26 + 7) = 31$; the other α_i need no clipping (each already satisfies all dense constraints). Objective: $(7 + 18 + 31 + 20) + (18 + 7 + 26 - 7) = 120$. Option A (clip the β side against the original α) gives $\hat{\beta} = (16, 7, 25, -7)$ with objective 119. Taking the max (Theorem 1),

$$\text{LB} = 120 \leq \text{OPT} = 121 \leq \text{UB} = 122,$$

so the pool solution is certified within $\frac{122-120}{122} = 1.64\%$ (width) of the dense optimum—here computed by hand; in the implementation it is two $O(n^2)$ min-passes. Note that the two one-sided options differ (120 vs 119), illustrating Remark 1.

A.3 Monotonicity of the upper bound in k

Since $E_k \subseteq E_{k'}$ for $k \leq k'$, every k -pool matching is a valid k' -pool candidate, hence $\text{UB}_{k'} \leq \text{UB}_k$; the pool gap is non-increasing in k by definition. The certified width is monotone in k on all 15 measured configurations (an empirical fact about the *repair*, not a theorem: the pool dual changes with k).

B Experimental details

B.1 Instance generators

- **uniform**: `rng = default_rng(seed)`; both sides = $100 \cdot \text{rng.random}((n, 2))$.
- **clustered**: centers $c_a = 100 \cdot \text{rng.random}((8, 2))$; each side assigns point i to center $a = i \bmod 8$ with $x_i = c_a + \mathcal{N}(0, 8^2 I)$; the target centers are $c'_a = c_a + \mathcal{N}(0, 3^2 I)$ with the same assignment (target-center jitter $\sigma = 3$, matching the description in §5).
- **1-D+noise**: $t_i \sim U[0, 100]$; $x_i = (t_i + \mathcal{N}(0, 2), \mathcal{N}(0, 2))$; independently $s_j \sim U[0, 100]$ from `default_rng(seed+1000)`; $y_j = (s_j + \mathcal{N}(0, 2), \mathcal{N}(0, 2))$.

Algorithm 1 Log-domain Sinkhorn (vanilla)

Require: cost $C \in \mathbb{R}^{n \times n}$, $\varepsilon > 0$, tolerance τ

```
1:  $u \leftarrow 0, v \leftarrow 0$ 
2: repeat
3:   for  $i = 1, \dots, n$  do
4:      $u_i \leftarrow u_i - \varepsilon \text{LSE}_j((v_j - C_{ij})/\varepsilon)$ 
5:   end for
6:   for  $j = 1, \dots, n$  do
7:      $v_j \leftarrow v_j - \varepsilon \text{LSE}_i((u_i - C_{ij})/\varepsilon)$ 
8:   end for
9:    $P_{ij} \leftarrow \exp((u_i + v_j - C_{ij})/\varepsilon)$ 
10: until  $\max_i |(P\mathbf{1})_i - 1| + \max_j |(P^\top \mathbf{1})_j - 1| \leq \tau$  or cost plateau
11: return  $P$ , value  $\langle C, P \rangle + \varepsilon \sum P_{ij} \log P_{ij}$ 
```

Algorithm 2 Greenkhorn (after [9, Alg. 4])

Require: $K = \exp(-C/\varepsilon)$ (unnormalized), potentials $x, y = 0$, residual metric $\rho(a, b) = b - a + a \log(a/b)$

```
1: maintain row/column marginals of  $A = D(e^x)KD(e^y)$  incrementally
2: repeat
3:    $I \leftarrow \arg \max_i \rho(1, (A\mathbf{1})_i)$ ;  $J \leftarrow \arg \max_j \rho(1, (\mathbf{1}^\top A)_j)$ 
4:   if  $\rho_I > \rho_J$  then
5:      $x_I \leftarrow x_I + \log(1/(A\mathbf{1})_I)$ 
6:   else
7:      $y_J \leftarrow y_J + \log(1/(\mathbf{1}^\top A)_J)$ 
8:   end if
9:   update  $A$  and the marginals of the touched line ( $O(n)$ )
10: until marginal tolerance or cost plateau
```

B.2 Algorithms

Hardening (`greedy_hard`): for $i = 1, \dots, n$, set $\pi(i) = \arg \max_j P_{ij}$ among unused columns, removing row i and column $\pi(i)$ from P ; the cost is $\sum_i C_{i, \pi(i)}$ (a feasible integer plan, hence $\geq \text{OPT}$).

Pool construction (`knn_pool`): for each source i , keep the k targets with smallest C_{ij} (and symmetrically for each target); the pool is the union. If the pool has no perfect matching (Hall's condition fails, e.g. $k = 1$), the solver reports infeasible.

Certificate (`pool_certified`): UB/perm/dual from `blossom_pool_duals`; LB_1 from Proposition 3 (max of the two one-sided repairs); $\text{LB}_2 = W(0.05 \overline{C})$ from one converged Sinkhorn run per configuration (shared across k); $\text{LB} = \max(\text{LB}_1, \text{LB}_2)$.

B.3 Parameters

Marginal tolerance 10^{-9} (application hook: 10^{-7} , the plan is hardened anyway); time caps 400s (dense grid), 240s (certificate study Sinkhorn runs); Blossom VI pool timeout 120s (never hit); costs rounded to milliunits for the integer solvers; all timings single-threaded wall time, same machine, same process order.

C Reproduction

All code, data, and results are open artifacts under the MIT license, available at <https://github.com/dimkadimon/OT-Blossom>. The repository layout:

ot-study/otlib.py	# sinkhorn, greenkhorn, exact, pool, # dual repair, entropic floor, certificate
ot-study/run_ot_study.py	# Part A/B tables -> results/ot_study.md
ot-study/run_greenkhorn_study.py	# A.2b, A.3, Part D -> results/ot_study.md
ot-study/run_pool_cert_study.py	# certificate study (15 x 3 cells) # -> results/pool_cert.md, asserts invariants
ot-study/app_hook_robots.py	# multi-robot study -> results/app_hook.md
paper/	# LaTeX source and figs for this paper

Each script caches its results (JSON) and re-renders in seconds; a full cold re-run takes ≈ 25 min (Greenkhorn grid), ≈ 20 min (certificate study), ≈ 6 min (application hook) on the testbed. The Blossom VI binary used for pool solves is available from its authors [6]; our driver uses the edge-list interface with the `-duals` flag.

D Prompts that directed this work

This appendix summarizes, in chronological order, the prompts used by D. K. to direct the work reported here. Prompts marked (verbatim) are quoted exactly; earlier prompts are paraphrased from the conversation record because their exact wording was not preserved.

1. (paraphrase) Opening question: can the Blossom (matching) algorithm be used for the TSP, and if so how—setting up the study of Edmonds’ algorithm as a component of TSP algorithms.
2. (paraphrase) Requests to implement and validate a Christofides heuristic using Blossom for the matching step against exact optima on small metric instances, and to benchmark the pre-shortcut multigraph.
3. (paraphrase) Request for a head-to-head comparison of Blossom V and Blossom VI (matching costs and wall times), including on dense complete graphs.
4. (verbatim, mid-task) “Actually you can now use Blossom VI” with a link to the Blossom VI repository—direction to make Blossom VI the default engine (retaining Blossom V for the head-to-head).
5. (paraphrase) Follow-up: “perturb the 2-matching, merge its cycles into one tour”—with the standing requirement that the merge be efficient and come close to the optimal tour, and that results be compared to state-of-the-art methods.
6. (paraphrase) Request for rigorous lower bounds via matching (2-factor / cycle-cover relaxations) to certificate the gap of the merge-based tours.
7. (verbatim) “Where else is Blossom useful?”—advisory question that produced the shortlist of exact-matching applications, of which optimal transport was ranked #1.
8. (verbatim) “do #1”—selection of the OT study as the next project.
9. (verbatim) “Yes build it. Continue clearing the publication bar”—approval to build the measured SOTA (Greenkhorn) baseline and the seed/distribution robustness sweep.

10. (verbatim) “Continue bridging the publication bar”—direction to complete the remaining two items: a pool solver with a gap certificate, and an application hook where the discrete plan is the deliverable.
11. (verbatim, this paper) “excellent! can you prepare an arXiv paper about it with references and proofs. Make it professional and publishable. Add an appendix section where you summarise the prompts I have used here. Can you add me to the author list (Dmitry Kamenetsky with email dkamenetsky@gmail.com) and the AI. Mention that all the work was done by AI with the suggestion from me. Also convert it to a pdf so I can look at it”